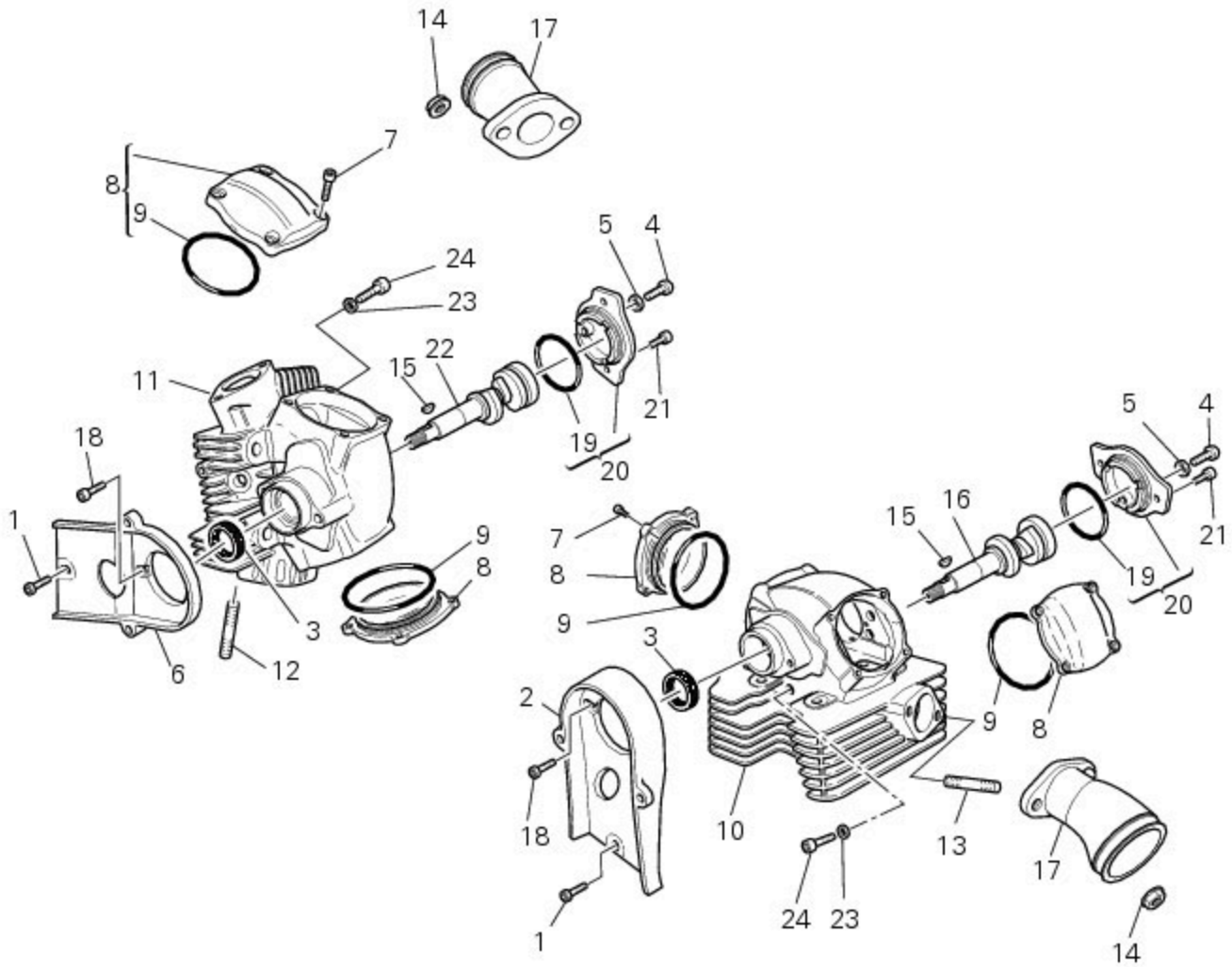


4.3 -Cylinder head assemblies: camshafts



- 1 Banjo bolt
- 2 Vertical cylinder belt inner cover
- 3 Sealing ring
- 4 Screw
- 5 Washer
- 6 Horizontal cylinder belt inner cover
- 7 Screw
- 8 Valve cover
- 9 O-ring
- 10 Vertical cylinder head
- 11 Horizontal cylinder head
- 12 Stud bolt
- 13 Stud bolt
- 14 Nut
- 15 Key
- 16 Camshaft-vertical cylinder head
- 17 Intake manifold
- 18 Screw
- 19 O-ring
- 20 Plug
- 21 Screw
- 22 Horizontal cylinder head camshaft
- 23 Copper gasket
- 24 Screw

1100 ABS	CYLINDER HEAD: TIMING SYSTEM
1100 ABS	VERTICAL CYLINDER HEAD
1100 ABS	HORIZONTAL CYLINDER HEAD
1100 ABS	INTAKE MANIFOLDS
1100 S ABS	CYLINDER HEAD: TIMING SYSTEM
1100 S ABS	VERTICAL CYLINDER HEAD
1100 S ABS	HORIZONTAL CYLINDER HEAD
1100 S ABS	INTAKE MANIFOLDS



Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the intake manifolds

Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings
Remove the fuel tank	L 2, Removal of the fuel tank
Remove the battery	P 2, Removal of the battery
Remove the battery mount	P 2, Battery support
Disconnect ECU connector from main wiring harness	M 3, Removal of the electronic control unit
Connect the starter motor/solenoid cable	P 3, Removal of the starter motor
Disconnect the HT leads	M 3, Spark plug
Remove the ABS control unit	G 9.2, Removing of the ABS control unit
Remove the airbox	L 7, Removal of the airbox
Disconnect the throttle cable	F 1, Removal of the throttle twistgrip
Remove the throttle body	L 6, Removal of the throttle body



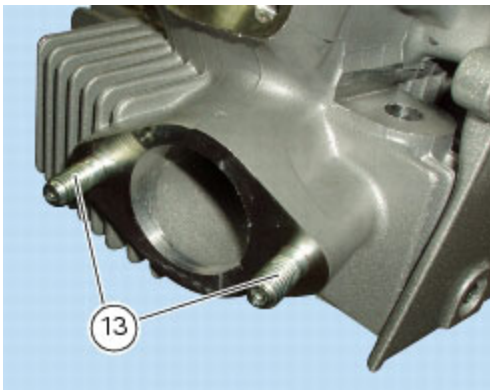
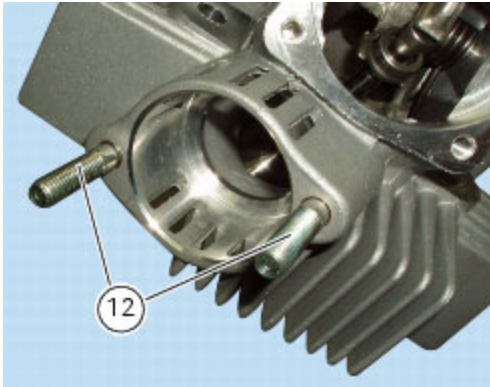
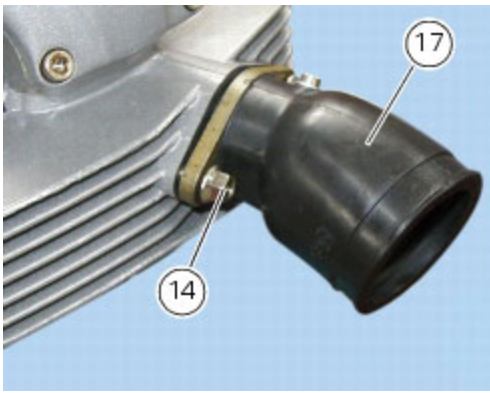
Note

The procedure can also be carried out with the engine installed in the frame. For clarity, the photos show the engine removed from the frame.

Unscrew and remove the nuts (14) securing the intake manifolds of the vertical and horizontal cylinder heads from the studs (13).

Remove the manifolds (17).

If necessary, remove the studs (12) and (13) from the cylinder heads.

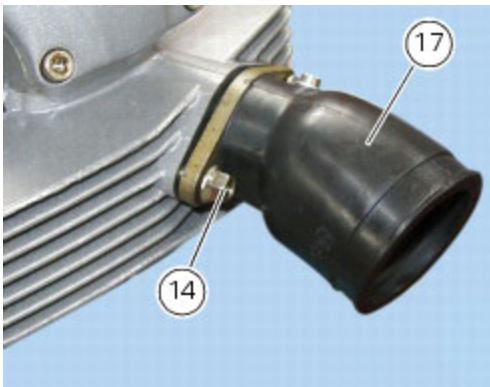


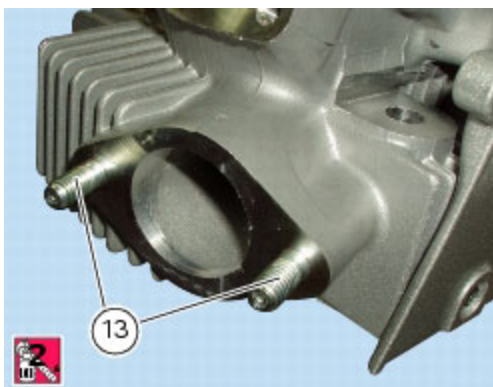
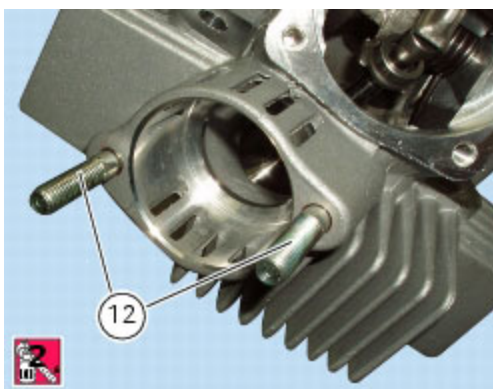
Refitting the intake manifolds

Apply the recommended threadlocker to the stud bolts (12) and (13) fit them on the cylinder heads and tighten to the specified torque (Sect. C 3, [Engine torque settings](#)).

Locate the intake manifolds (17) and secure with the nuts (14).

Tighten the nuts (14) to the specified torque (Sect. C 3, [Engine torque settings](#)).





Operation	Section reference
Refit the throttle body	L 6, Refitting the throttle body
Connect the throttle cable	F 1, Refitting the throttle twistgrip
Refit the airbox	L 7, Refitting the airbox
Refit the ABS control unit	G 9.2, Refitting the ABS control unit
Reconnect the HT leads	M 3, Spark plug
Connect the starter motor/solenoid cable	P 3, Refitting the starter motor
Connect ECU connector to main wiring harness	M 3, Refitting the electronic control unit
Refit the battery mount	P 2, Battery support
Refit the battery	P 2, Refitting the battery
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat

Removal of the side covers and the timing belt inner covers

Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings

Remove the fuel tank	L 2, Removal of the fuel tank
Remove the horizontal cylinder exhaust pipe	L 8, Removal of the exhaust system
Disconnect the timing-side spark plug cables	M 3, Spark plug
Remove the side timing belt covers	N 4.2, Removal of the timing belt covers
Remove the timing belts and camshaft timing belt pulleys on the heads	N 4.2, Removal of the timing system assembly



Note

For clarity, the figures show the cylinder head removed from engine.

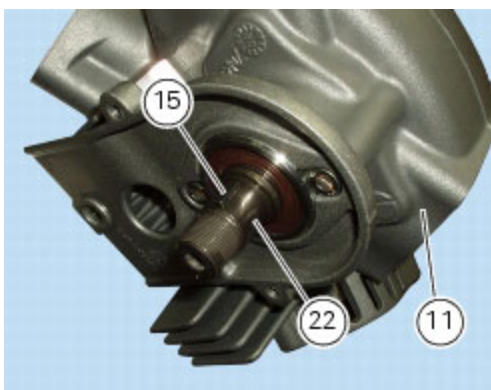
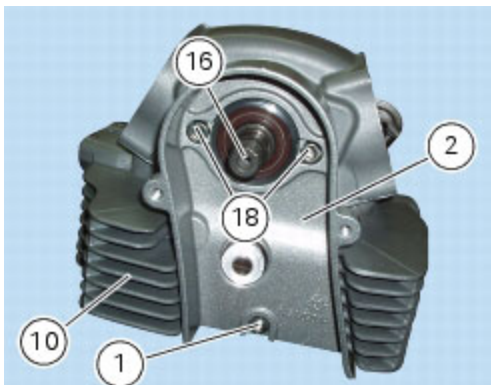
Remove the retaining screws (18) and (1) and remove the vertical belt inner cover (2) from the vertical cylinder head (10).

Repeat the operation for the horizontal cylinder head (**11**).

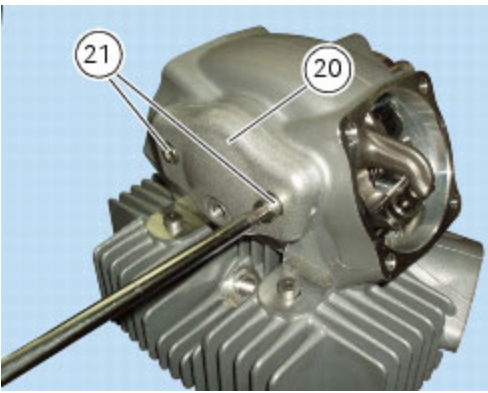


Note

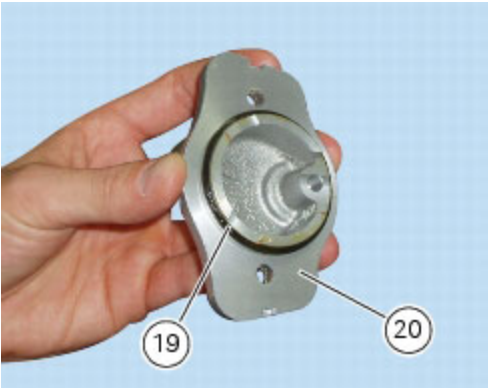
When removing the covers, take care not lose the tongues (15) located on camshafts (16) and (**22**).



Unscrew the screws (21) of the cover (20) of the camshaft (**16**) and (22).

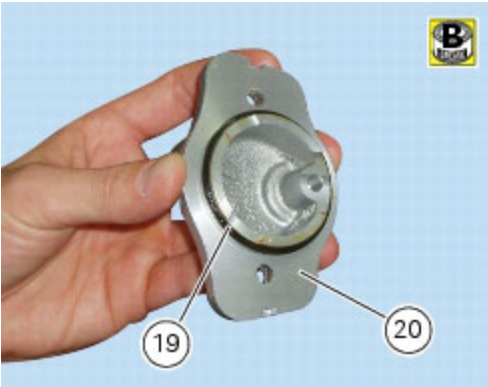


Remove the camshaft end cap (20) from the cylinder head and recover the O-ring (19).

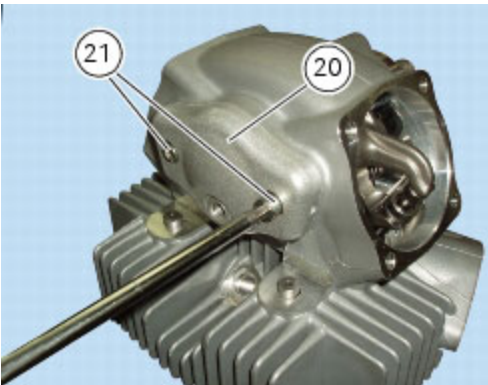


Refitting of the side covers and the timing belt inner covers

Lubricate the new O-rings (19) with the recommended grease and fit on the end cap (20).
Fit the end cap (20) on the left-hand side of the cylinder head.



Secure the end cap (20) with the screws (21).
Tighten the screws (21) to the specified torque (Sect. C 3, [Engine torque settings](#)).

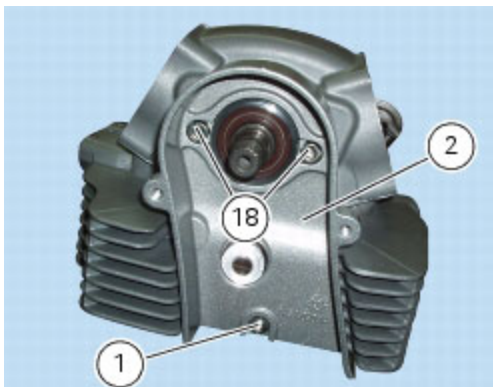


Note

Take care not to dislodge the key (15) located on camshaft.



Locate the vertical belt inner cover (2) on the left-hand side of the cylinder head and insert the retaining screws (1). Take care to position special screw (1) correctly.
Tighten the screws (18) and (1) to the specified torque (Sect. C 3, [Engine torque settings](#)).



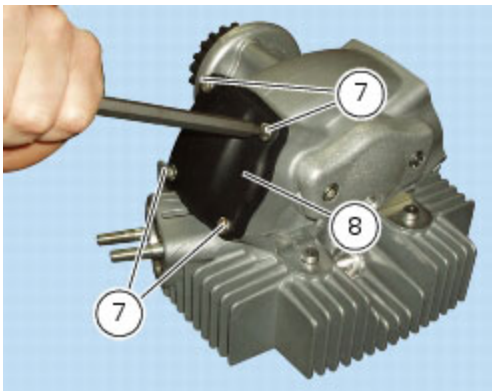
Repeat the above operations for the other cylinder head.

Operation	Section reference
Refit the camshaft timing belt pulleys on the heads and timing belts	N 4.2, Refitting the timing system assembly
Refit the timing belt covers	N 4.2, Refitting the timing belt covers
Connect the timing-side spark plug cables	M 3, Spark plug
Refit the horizontal cylinder exhaust pipe	L 8, Refitting the exhaust system
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat

Removal of the valve covers

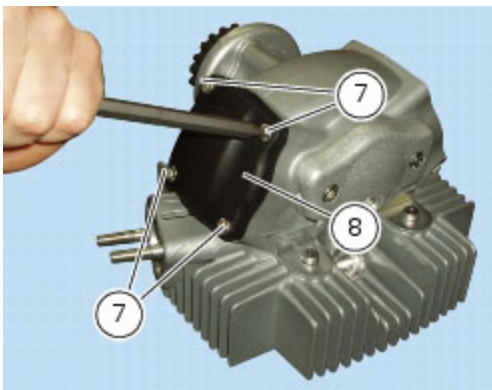
Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings
Remove the fuel tank	L 2, Removal of the fuel tank
Remove the oil cooler from the frame	N 2.2, Removal of the oil cooler

Undo the retaining screws (7) of valve covers (8) and remove the covers with relative O-rings (9).



Refitting the valve covers

Always fit new O-rings (9) when refitting.
 Apply the recommended grease to the seals (9).
 Fix the valve covers (8) with the relative O-ring (9) to the cylinder head.
 Tighten the screws (7) to the specified torque (Sect. C 3, [Engine torque settings](#)).



Operation	Section reference
Refit the oil cooler to the frame	N 2.2, Refitting the oil cooler
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat

Removal of the camshafts

Operation	Section reference
Remove the seat	E 3, Removal of the seat
Remove the tank covers	E 2, Removal of the fuel tank fairings

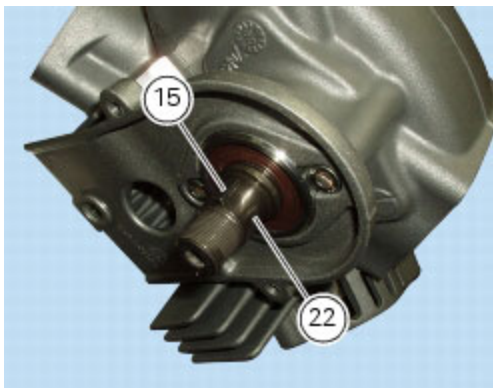
Remove the fuel tank	L 2, Removal of the fuel tank
Remove the horizontal cylinder exhaust pipe	L 8, Removal of the exhaust system
Disconnect the timing-side spark plug cables	M 3, Spark plug
Remove the side timing belt covers	N 4.2, Removal of the timing belt covers
Remove the timing belts and camshaft timing belt pulleys on the heads	N 4.2, Removal of the timing system assembly
Remove the side covers	N 4.3, Removal of the side covers and the timing belt inner covers



Note

This operation can be carried out with the rocker arms installed in the cylinder head. For clarity, the photos show the cylinder head with the rocker arms removed.

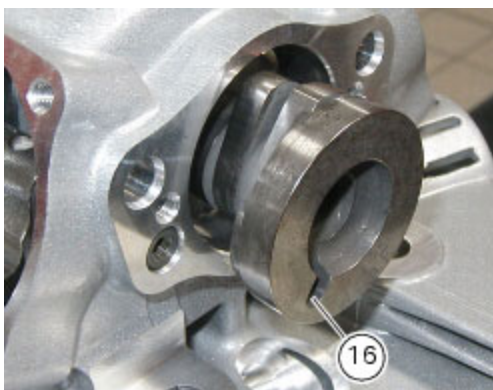
Remove the key (15) from the camshaft.

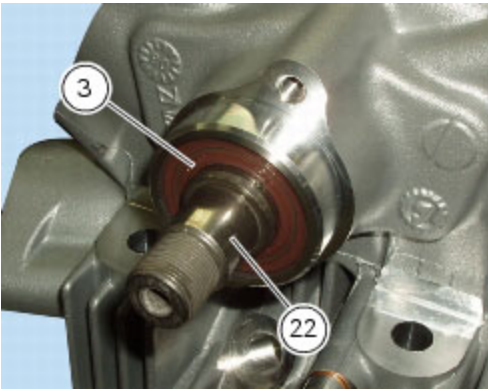


Note

To prevent possible damage to components, turn the camshaft (16) or (22) through a series of rotations so that the cams do not foul the rockers.

Withdraw the camshaft (16) or (22) from its housing.
Remove the oil sealing ring (3) from the cylinder head.





Inspecting the camshafts

Check the cam contact surfaces for scratches, grooves, steps and waving. Worn cams are frequently the cause of poor timing, which leads to loss of engine power.

Install the camshaft between two centres and check for distortion using a dial gauge:

- service limit:

0.1 mm.



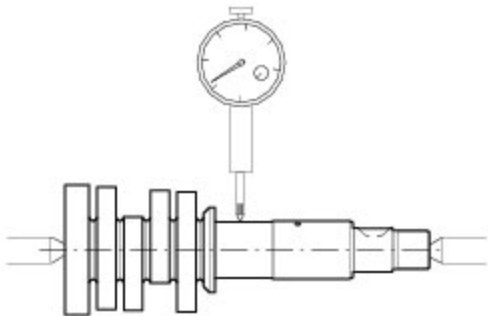
Note

The effective distortion will be half the total reading obtained on the dial gauge.

Each camshaft is stamped with an identification letter:

O: horizontal

V: vertical.

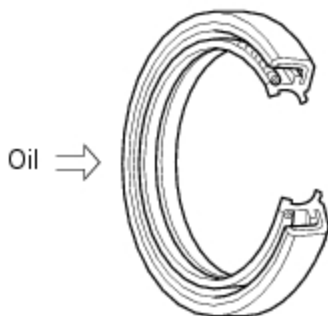


Checking the oil seals

Fit new oil seals at each engine overhaul.

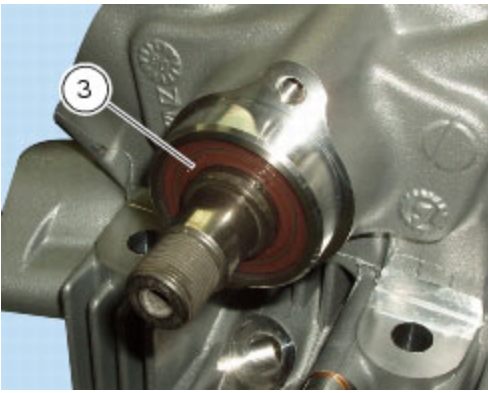
Insert the new oil seals squarely in their bores. Use suitable drifts.

Lubricate the lip of oil seals after assembly. This operation must be performed very carefully.



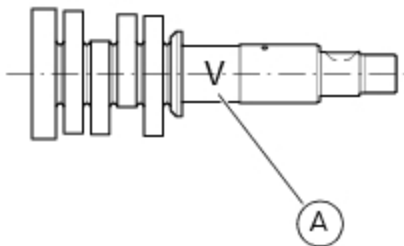
Refitting the camshafts

Every time you reassemble, fit a new seal (3) in the seat on head right-hand side, setting it with the side with no spring as shown below.

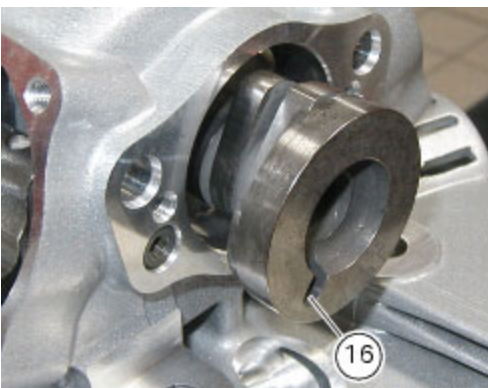
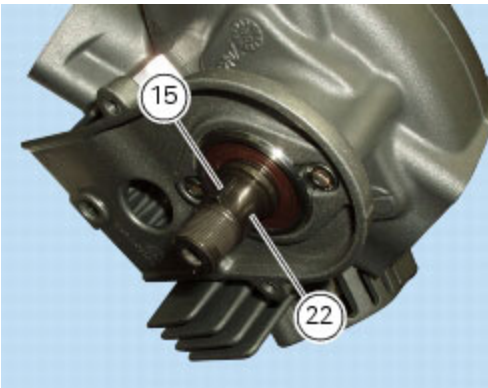


Important

If the valve rocker arms are already installed in the cylinder head (Sect. N 4.4, [Refitting of the rocker arms, shims and valves](#)), turn the camshaft (16), through a series of rotations, so that the cams do not foul the rocker arms. Before fitting the camshaft, check the mark (A): "V" means vertical, "O" means horizontal.



Apply graphite oil to the seats of the camshaft so that it is lubricated also during the initial revs of the engine. Install the camshaft (16) or (22) in the cylinder head. Check the condition of the tongue (15) and its groove, and install it.



Operation	Section reference
Refit the inner timing belt covers	N 4.3, Refitting of the side covers and the timing belt inner covers
Refit the camshaft timing belt pulleys on the heads and timing belts	N 4.2, Refitting the timing system assembly
Refit the timing belt covers	N 4.2, Refitting the timing belt covers
Connect the timing-side spark plug cables	M 3, Spark plug
Refit the horizontal cylinder exhaust pipe	L 8, Refitting the exhaust system
Refit the fuel tank	L 2, Refitting the fuel tank
Refit the tank covers	E 2, Refitting the fuel tank fairings
Refit the seat	E 3, Refitting the seat